

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Case Study

COURSE NAME: Cloud Computing and Big Data Analytics

COURSE CODE: CSA15

COURSE OUTCOME COVERED

CO3: Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications.
(BL3)

Dave's Taxonomy: Precision (Level 3)

Question Count: 3

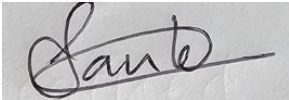
Total Marks: 30

CODE OF CONDUCT

I SANDEEP K / 192521253 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



SANDEEP K / 192521253

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1. Case Study 1: Smart Retail Inventory Management

The cloud system connects store devices and dashboards to centralized services. Cloud storage and analytics support inventory visibility and demand forecasting.

- POS and dashboards communicate through APIs.
- Store products, transactions and analytics datasets in cloud storage.
- Use platform services for analytics and forecasting.
- Protect access with VPNs, firewalls and API gateways.
- Combine infrastructure, services, APIs, storage and analytics.

Cloud Case-Study Architecture

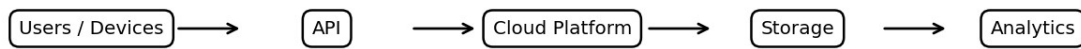


2. Case Study 2: Cloud-Based Document Collaboration

The collaboration system requires concurrent access, synchronization and version control. Distributed cloud storage and APIs support multi-device usage.

- Users edit and share documents through browsers.
- Store documents in distributed cloud storage.
- Maintain version control.
- Use APIs for synchronization.
- Use encryption, identity management, permissions and reliable networking.

Cloud Case-Study Architecture



3. Case Study 3: Online Food Ordering Platform

The platform combines web/mobile clients with REST APIs and cloud infrastructure. Managed infrastructure provides scalability while security controls protect users.

- Clients communicate with backend services through REST APIs.
- Store menu images, invoices and customer data securely.
- Use VMs and managed Kubernetes for scalability.
- Use authentication tokens and HTTPS.
- Apply role-based access control.

Cloud Case-Study Architecture

